Standard Operating Procedure: Refinery HSE & Site Protective Protocols

1. Administrative Scope and Safety Objective

This standard operating procedure establishes mandatory health, safety, and environmental (HSE) regulations for all personnel, process operators, maintenance contractors, and inspection engineers within the battery limits of Mangalore Refinery and Petrochemicals Limited (MRPL). Compliance with OISD-105 and OISD-154 statutory safety frameworks is mandatory. Zero tolerance is enforced for non-compliance.

2. Mandatory Personal Protective Equipment (PPE) Matrix

All personnel entering process operating areas must continuously wear certified PPE including:
a) Flame-Retardant Clothing (FRC) meeting NFPA 2112 and ISO 11612 standards with antistatic weave.
b) Portable Multi-Gas Detector (LEL, O2, CO, H2S) attached inside the breathing zone (within 25 cm of mouth/nose). The H2S alarm threshold is strictly set at: Low Alarm = 10 ppm, High Alarm = 15 ppm.
c) Heavy industrial safety footwear compliant with EN ISO 20345 S3 (steel toe, oil-resistant, antistatic sole).
d) Impact-resistant safety eyewear conforming to ANSI Z87.1 with permanent side shields.
e) Hearing protection mandatory in all designated high-noise compressor and pump zones (>85 dBA).

3. Hazardous Area Classification and Permit-To-Work (PTW)

Operational plant areas are zoned according to hazardous combustible atmosphere likelihood:
- Zone 0: Explosive hydrocarbon atmosphere continuously present (>1000 hours/year). No hot work allowed.
- Zone 1: Explosive atmosphere likely to occur under normal operating conditions (10 to 1000 hours/year).
- Zone 2: Explosive atmosphere not likely in normal operation and only for short durations (<10 hours/year).

Hot Work Permits (HWP) require authorized signature from Shift In-Charge and continuous LEL explosive gas monitoring. If combustible gas concentration reaches 1% LEL, hot work must halt immediately. A dedicated fire watch equipped with a 50 kg Dry Chemical Powder (DCP) wheeled unit must stand by at all times.

4. Hydrogen Sulfide (H2S) Emergency Action Protocol

In the event of an audible H2S toxic gas alarm sounding from fixed site beacons or personal monitors:
1. Check the nearest wind sock for wind direction immediately.
2. Evacuate cross-wind or up-wind to the nearest designated assembly point (Muster Station Alpha or Bravo).
3. Never walk downwind or into low-lying drainage pits or pipe trenches.
4. Certified rescue teams must don 30-minute 300-bar Self-Contained Breathing Apparatus (SCBA) before entry.
5. Notify Main Control Room (MCR) via plant emergency phone extension 5555 or UHF Channel 1.